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THE COMPUTATION OF RADIATION FROM NONEQUILIBRIUM HYPERSONIC FLOWS

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Abstract

The results of the solution of the equations that describe a hypersonic ionized flow about an elliptically blunted cone are presented. The flow conditions correspond to those of the proposed Aeroassist Flight Experiment (AFE) vehicle at altitudes between the perigee at 78 km and the approximate limit of the continuum regime at 90 km. For the free-stream velocities of interest, about 9 km/sec, the flowfield is out of thermo-chemical equilibrium, electronically excited, ionized and radiating. The gas consists of eight chemical species including free electrons. The thermal state of the gas is modeled with a translational-rotational temperature, four vibrational temperatures for the diatomic species and an electron-electronic temperature. The electronic excitation of molecules is included. The nonequilibrium air radiation from each fluid element is computed and the radiative heat flux at the body surface is determined. The stagnation point radiative heating result agrees with previous calculations.

Nomenclature

c_s = average thermal speed of species s
c_{vs} = translational-rotational specific heat
E = total energy per unit volume
E_e = electron energy per unit volume
E_{vs} = vibrational energy per unit volume
$e_{el\,s}$ = electronic energy per unit mass
e_{vs} = vibrational energy per unit mass
e^*_{vs} = equilibrium vibrational energy per mass
g_{is} = degeneracy of electronic level i, species s
h°_s = heat of formation of species s
$k_f,\ k_b$ = forward and backward reaction rates
K_{eq} = equilibrium reaction rate
M_s = atomic mass of species s
$\mathcal{M}$ = Mach number
N_s = number density of species s
q_{rad} = radiative heat flux
Q_{rad} = radiative intensity
Q_{T-vs} = translation-vibration energy transfer rate
Q_{e-vs} = electron-vibration energy transfer rate
R = universal gas constant
S_s = exponent for use in diffusion model
T = translational-rotational temperature
T_{vs} = vibrational temperature of species s
T_e = electron-electronic temperature
u_i = i direction velocity
ρ_s = density of species s
ξ = axial distance from nose
η = normal distance from wall
$\eta_r,\ \theta_r$ = constants in Arrhenius reaction rate
θ_{is} = electronic energy of level i, species s
θ_s = characteristic energy for diffusion model
σ_s = collision cross-section
τ_{cs} = collision limited relaxation time
τ_{es} = electron-vibration relaxation time
τ_{L-Ts} = Landau-Teller relaxation time
τ_{vs} = translation-vibration relaxation time

Introduction

Several conceptual designs for vehicles that would maneuver in the earth's atmosphere have been proposed recently[1,2]. In particular, the Aeroassisted Orbital Transfer Vehicle (AOTV) would be used to transfer payloads between low and high earth orbits. During return from high earth orbit, it would fly into the atmosphere and, using aerodynamic drag, slow down to low earth orbit speed. This aerobraking has the advantage of reducing the amount of fuel required for an orbital transfer. The vehicle would fly in the upper atmosphere where, because of the low density, the flow is significantly removed from thermo-chemical equilibrium. The degree of nonequilibrium affects the aerodynamic forces and convective heating acting on the vehicle. And in the AOTV flight regime, the chemical and thermal state of the flowfield has a large influence on the radiative heating.

The nature of this chemically reacting and highly thermally excited flowfield is described by the Navier-Stokes equations expanded to include a mass conservation equation for each chemical species and an equation for the conservation of each of the primary energy modes of the gas. A recent paper[3] describes the solution of these equations to determine the electron concentrations about a sphere-cone travelling at hypersonic speeds. The gas was modeled as composed of seven different chemical species and

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characterized by a translational-rotational temperature, a vibrational temperature for each diatomic species, and an electron-electronic temperature. However, the model used to describe the thermal state of the gas had several inadequacies. In this study, improved models for the vibration-translation and electron-vibration energy exchange mechanisms are implemented. Electronic excitation of molecules is also included. The results of the flowfield computations are used to determine the radiative heat flux to the surface of the body using a previously reported technique[4].

The test cases that are considered are for the proposed Aeroassist Flight Experiment (AFE) for the part of the trajectory where the vehicle is in the continuum regime. The current design for this aerobrake has an elliptically blunted cone forebody and a skirt-type afterbody. An axisymmetric approximation to this body has been presented which is made up of an ellipsoid with a 2.3 m nose radius and a 60° half-angle cone[5,6]. Results are presented for the perigee conditions of the AFE vehicle at 78 km and at the approximate limit of the continuum regime at 90 km.

Governing Equations

The partial differential equations that describe a continuum chemically reacting, ionizing, and thermally relaxing flow have been discussed previously[3,7]. A mass conservation equation is required for each chemical species that exists in the flowfield. A momentum conservation equation is needed for each spatial dimension. Conservation equations for the vibrational energy of each diatomic species and electron energy are included and a total energy conservation equation is required. Here we consider a two-dimensional flowfield made up of the seven primary chemical species found in high temperature air. These constituents are N_2, O_2, NO, NO^+, N, O, and e^-. Since four of these molecules are diatomic, four separate vibrational energy equations are required. This brings the total number of equations to fifteen when the electron and total energy equations are included. Thus, there are six temperatures that characterize the thermal state of the gas. These are T, the translational-rotational temperature, T_{vs}, the four species vibrational temperatures, and T_e, the electron-electronic temperature. The concentration of the eighth chemical species, N_2^+, is computed from the flowfield solution with a technique discussed below. A description of the modifications that have been made to the previously presented gas model follows.

The equation set is solved in a fully coupled fashion using an implicit time-marching technique developed by MacCormack[8]. It uses Gauss-Seidel line-relaxation to reduce the cost of using an implicit method and it employs flux-splitting to maintain numerical stability in subsonic and supersonic regions of the flow. The technique has been shown to be applicable to chemically reacting flows[3,9].

Equations of State

The equation of state for the multi-temperature reacting gas is derived by subtracting the separate energy modes from the total energy, which yields

$$E = \sum_{s \neq e} \rho_s c_{vs} T + \frac{1}{2} \sum_{s \neq e} \rho_s u_i u_i + \sum_{s=\mathrm{vib}} E_{vs} + E_e + \sum_{s \neq e} \rho_s h_s^\circ + \sum_{s \neq e} \rho_s e_{\mathrm{el}\,s}, \tag{1}$$

$$E_e = \rho_e c_{ve} T_e + \frac{1}{2} \rho_e u_i u_i.$$

The electronic energy per unit mass, $e_{\mathrm{el}\,s}$, which is governed by the electron temperature, is given by

$$e_{\mathrm{el}\,s} = \frac{R}{M_s} \frac{\sum_{i=1}^{\infty} g_{is} \theta_{is} \exp(-\theta_{is}/T_e)}{\sum_{i=0}^{\infty} g_{is} \exp(-\theta_{is}/T_e)}. \tag{2}$$

The terms used here are described in the nomenclature section. In the present study, only the terms in the sum up to $i = 1$ are included. Implicit in this formulation is that the translational temperature of the free electrons is in equilibrium with the temperature that characterizes the bound electrons associated with the molecules. This assumption is based on the fact that there is essentially no barrier to bound-free electron exchanges, and therefore the temperatures that characterize each mode must be nearly equal.

Energy Exchange Mechanisms

The thermal state of the nonequilibrium gas is described by, in this case, six separate temperatures. The temperature that characterizes the energy in each mode relaxes toward a common equilibrium temperature at a finite rate.

The rate of energy exchange between vibrational and translational modes has been discussed by a number of authors[10–13]. The rate of change in the population of the vibrational states at low temperatures is described well by an equation of the Landau-Teller form[10]. However, at high vibrational temperatures the exchange process is primarily governed by a comparatively slow diffusive process[11–13]. A bridging function between the faster Landau-Teller relaxation rate and the slower diffusive rate has been proposed[11–13]. The resulting expression for the translational-vibrational exchange term is

$$Q_{T-vs} = \rho_s \frac{e^*_{vs}(T) - e_{vs}}{\tau_{vs}} \left| \frac{T_{shk} - T_{vs}}{T_{shk} - T_{vs\,shk}} \right|^{S_s - 1}, \quad (3)$$

where the bridging function, S_s, is given by

$$S_s = 3.5 \exp(-\theta_s / T_{shk}). \quad (4)$$

The quantities T_{shk} and $T_{vs\,shk}$ are the translational-rotational and species vibrational temperatures evaluated just behind the bow shock wave. For a general location in the flowfield, the streamline that passes through that point should be traced back to the shock wave where T_{shk} and $T_{vs\,shk}$ are evaluated. This process is not easy to generalize, and as an approximation for the two-dimensional flowfields discussed here, all post-shock temperatures were evaluated at the nearest shock wave location. For these AFE forebody flowfields this approximation is reasonable, however a better technique is needed for this procedure. The characteristic temperatures, θ_s, are listed in the Appendix.

The relaxation time, τ_{vs}, is a combination of the Landau-Teller rate expression[10] and the limiting cross-section rate equation[12,13] such that

$$\tau_{vs} = \tau_{L-Ts} + \tau_{cs}, \quad (5)$$

where

$$\tau_{cs} = \frac{1}{c_s \sigma_v N_s}. \quad (6)$$

c_s is the average molecular speed of species s, $c_s = \sqrt{8RT/\pi M_s}$ and N_s is the number density of the colliding particles. The expression for the limiting collision cross-section, σ_v, is assumed to be given by[12]

$$\sigma_v = 10^{-21}(50{,}000/T)^2 \quad \mathrm{m}^2. \quad (7)$$

The use of this expression for the vibrational relaxation rate considerably reduces the peak vibrational temperature compared to that of the Landau-Teller expression, and hence has a large influence on the quantity of radiation calculated to be emitted from the flowfield.

The coupling of the electron temperature with the vibrational temperature of nitrogen is strong. We assume that the rate of exchange between vibration and electron energy modes has the form

$$Q_{e-vs} = \rho_e \frac{M_s}{M_e} \frac{e^*_{vs}(T_e) - e_{vs}}{\tau_{es}}, \quad \text{for } s = N_2. \quad (8)$$

Lee[14] derived an expression for the relaxation time of electron-vibration energy transfer, τ_{es}, by solving the system of master equations that accounts for transitions of multiple levels. τ_{es} was curve-fit to the results of Lee using two quadratics in the logarithm (base 10) of the electron temperature. This expression is

$$\log(p_e \tau_{es}) = \begin{cases} 7.50(\log T_e)^2 - 57.0 \log T_e + 98.70, & T_e < 7000 \text{ K} \\ 2.36(\log T_e)^2 - 17.9 \log T_e + 24.35, & T_e > 7000 \text{ K} \end{cases} \quad (9)$$

where p_e is the electron partial pressure in atmospheres and the resulting τ_{es} is in seconds. The relaxation time is smallest at $T_e = 7000$ K, due to a strong resonance between the electron and vibration modes. τ_{es} increases rapidly on either side of this temperature.

Reaction Rates

The chemical reactions between the components of the gas have been assumed to be described by rate equations of the Arrhenius form. The two-temperature model for the reaction rates developed by Park[11,12] (the TT_v model) was extended to determine the temperature that governs these rates. For instance, the dissociation reaction rate of a species depends on the geometric mean of its translational and vibrational temperature, $\sqrt{TT_{vs}}$. Using this model, instead of the standard model governed by the translational temperature alone, leads to a lower the rate of dissociation and suppresses the degree of reaction of the gas to more accurate levels.

N_2^+ Concentration Approximation

The concentration of the diatomic nitrogen ion, N_2^+, which may be significant in high temperature flows, may be approximated from the computed flowfield using a quasi-steady-state (QSS) approach. This technique has been used with moderate success to compute electron number densities[3], however the technique used in this study is somewhat different. The present approach assumes that the N_2^+ producing reaction is in equilibrium at the local thermochemical conditions. This reaction is

$$N + N \rightleftharpoons N_2^+ + e^-, \quad (10)$$

and in equilibrium we have

$$K_{eq} = \frac{k_f(T)}{k_b(\bar{T})} = \frac{N_{N_2^+} N_{e^-}}{N_N^2}, \qquad (11)$$

where the forward rate constant has been assumed to be a function of T alone, while the backward rate constant has been assumed to be characterized by the average temperature, $\bar{T} = \sqrt{T_{vN_2} T_e}$. The number density of electrons is the sum of those in the previously computed flowfield (denoted by $N_{e^-}^*$) and those produced by reaction (10), which will be equal to $N_{N_2^+}$. Therefore, using (11) we can write

$$\begin{aligned} N_{N_2^+}(N_{N_2^+} + N_{e^-}^*) &= N_N^2 k_f(T) \frac{K_{eq}(\bar{T})}{k_f(\bar{T})} \\ &= N_N^2 \left(\frac{T}{\bar{T}}\right)^{\eta_r} \exp\left(\frac{\theta_r}{\bar{T}} - \frac{\theta_r}{T}\right) K_{eq}(\bar{T}), \end{aligned} \qquad (12)$$

where the Arrhenius form of the forward rate (with constants η_r and θ_r) and a curve-fit expression for the equilibrium expression are used.

The QSS assumption will give good results in the regions where the gas is near thermo-chemical equilibrium, but will over-predict the number of N_2^+ ions where the gas is undergoing compression and is significantly nonequilibrium. Thus this method gives an upper bound on the ion concentration in the forebody flowfield.

Radiation Calculations

The computation of the radiative emission from the flowfield was performed by first obtaining a converged result for the case of interest. Then these results were fed into the NEQAIR algorithm of Park[4] which calculates the total radiative power emitted from the gas between wavelengths of $0.2\ \mu m$ and $1.5\ \mu m$. This code calculates the number densities of various electronic states by invoking the so-called quasi-steady-state condition for the conservation of electronic energy states. This analysis is based on the assumption that the rates for populating and depopulating an electronic state are much larger than the difference between them, which is the rate of change of the state population. The rates for population and depopulation are known functions of the thermodynamic state of the gas.

The gas is assumed to be optically thin above $0.2\ \mu m$ and optically opaque for wavelengths below $0.2\ \mu m$. Also it is assumed that the amount of radiative emission is small relative to the total energy content of the gas. Thus, we do not have to couple the radiative heat loss to the flowfield calculation. The so-called tangent-slab approximation is also made. With this simplification, we assume that the radiative flux at a point on the body is one-half of the integral of the total radiative intensity in the gas normal to the wall. This may be written as

$$\begin{aligned} q_{rad} &= \frac{1}{2}\int_0^\infty Q_{rad}(\eta)\, d\eta, \\ q_{rad\ i} &\simeq \frac{1}{2}\sum_j Q_{rad\ i,j}\left(\eta_{i,j+\frac{1}{2}} - \eta_{i,j-\frac{1}{2}}\right), \end{aligned} \qquad (13)$$

where Q_{rad} has units of power per volume.

Results

The results are preliminary in nature and should not be used for design or policy decisions. The numerical method has been compared to the RAM-C II experiment[3]. However, the physical model of the gas has not been compared to experiment for AFE flight conditions. As such, the results must be used with caution.

The flowfield around the axisymmetric version of the AFE vehicle was computed for two conditions in the proposed trajectory. The first is for the perigee conditions at 78 km and a free-stream velocity of 8.91 km/sec. The second is the case at 90 km and 9.89 km/sec which is the approximate limit of the continuum regime. Both cases were run with an assumed fixed wall temperature of 1000 K and a non-catalytic wall. Table I gives the free-stream conditions for each case. The composition of the air at 78 km was assumed to be 79% N_2 and 21% O_2 and at 90 km to be 76.6% N_2, 23.23% O_2 and 0.17% O (by mass). A plot of the grid used for the low altitude case is presented in Figure 1.

Table I. Free-stream Conditions for AFE Test Cases.

	78km	90km
u_∞ (m/s)	8910	9890
T_∞ (K)	197	188
ρ_∞ (kg/m^3)	2.78×10^{-5}	3.14×10^{-6}
Re	43400	5670
$\mathcal{M}_\infty$	31.6	36.4

Results for AFE at 78 km

The perigee conditions of the AFE flight trajectory will expose the vehicle to the highest radiative and convective heating and the largest drag forces. The accurate prediction of these aerothermal loads is critical to the design of the AFE. A discussion of the nature of the flowfield and the surface heating follows.

Figure 2, shows the density distribution along the stagnation streamline. The shock layer and the boundary layer are distinct in this case. If the standoff distance is defined to be the point where the density rise is six times the free-stream density, the shock standoff is 0.137 m, or 5.96% of the nose radius. Figure 3 is a plot of the molar concentrations along the stagnation streamline including the QSS approximation for N_2^+. The peak molar concentration of NO^+ ions is 1.83×10^{-3} and the peak concentration of N_2^+ is slightly higher. The SPRAP program of Park[12] was also used to compute the stagnation streamline conditions for this case. The peak NO^+ concentration from the current results is nearly three times higher than the SPRAP results. However, there are significant differences in the treatment of the shock wave and the rate constants used in the two methods. A previous study using the current method has shown excellent agreement with experiment for electron number densities[3] and this is a justification of the results obtained with the present method. The qualitative nature of the N_2^+ concentration distribution and the ratio between the concentration of NO^+ and N_2^+ are very similar for both methods.

The translational-rotational, nitrogen vibrational, and electron-electronic temperature distributions on the stagnation streamline are plotted in Figure 4. The other three vibrational temperatures are not plotted, because they have a similar behavior as that of nitrogen. There is a strong degree of thermal nonequilibrium near the shock, but for this case, the electron and vibrational temperatures equilibrate with the translational-rotational temperature within about 50% of the distance to the wall. The electron temperature rises near the shock due to translation-electron exchanges, and rises further downstream as the vibrational temperature becomes excited and the electron-vibration exchanges become more efficient.

Figure 5 is a plot of the radiative emission power along the stagnation streamline as computed by the NEQAIR program. The peak value of $2.63\ W/cm^3$ at $\eta = 0.0816$ m corresponds to the region where the vibrational and electron temperatures are largest.

Figures 6a and 6b show the calculated convective and radiative heating to the body surface as a function of distance from the nose. The stagnation point convective heating value is $16.5\ W/cm^2$. The heating rises on the ellipsoidal part of the body to a peak of $19.3\ W/cm^2$ and then drops to a nearly constant $14\ W/cm^2$ for the length of the cone. The rise in the heat transfer away from the nose is postulated to be a result of the ellisoidal surface curvature. The gas accelerates around the ellipsoidal nose, causing the boundary layer to thin, and increasing the temperature gradients and the heat transfer. The surface pressure distribution also has an interesting behaviour, as shown in Figure 6c. The gas experiences an adverse pressure gradient after the ellipsoid-cone juncture at $\xi = 0.9$ m. This is a result of centrifugal forces decreasing the pressure on the ellipsoidal surface as compared to the adjacent conical surface. These effects have been observed experimentally[15]. The stagnation point convective heating is similar to the $15.2\ W/cm^2$ predicted by Fay and Riddell with Goulard's correction for a non-catalytic wall[16,17], but it is considerably lower than the $24.8\ W/cm^2$ that Moss *et al.*[6] calculated. This discrepancy is probably due to the inclusion of a catalytic wall boundary condition in their study. The influence of a catalytic wall on the convective heat transfer can be profound for cases where the flowfield is highly reactive[18].

The peak radiative heating for this case occurs at the stagnation point and is computed to be $5.94\ W/cm^2$. This decreases rapidly and reaches a fairly constant value of about $2\ W/cm^2$ on the conical surface. The stagnation point radiative heat transfer agrees very well with the $6.63\ W/cm^2$ result of Park[12].

The next figures give a more qualitative description of the flowfield in the form of contour plots. Figure 7 is a plot of the percent N_2 mass concentration and shows that the gas is most highly reacted near the stagnation point and the degree of reaction decreases with axial distance. Figure 8 plots the contours of translational-rotational temperature. A comparison with Figure 9, which presents N_2 vibrational temperature contours, shows that these two temperatures are nearly in equilibrium for a large part of the flowfield. However, Figure 10 shows that the electron temperature remains lower than both other temperatures.

Figure 11 is a contour plot of the radiative emission power from the flowfield. The largest emission occurs on the stagnation streamline where the temperatures and the density are highest. The radiative power falls off rapidly from this point to a fairly constant level on the conical section of the body.

In summary, the results from this case indicate that the flowfield is highly reactive and radiative. The stagnation point heat transfer results are consistent with previously published experimental work. A major portion (26%) of the total heat transfer to the stagnation point is from radiation for this non-catalytic wall case. The maximum convective heat transfer occurs off of the nose, near the ellipsoid-cone juncture. The inclusion of thermal nonequilibrium for this case is mandatory for the accurate calculation of radiation because approximately a 5% error in the vibrational and electron temperatures results in a factor of two error in the radiative emission. (This can be shown by running the NEQAIR program with T_{ve} and T_e 5% larger than they are computed to be.)

Results for AFE at 90 km

The next case is the axisymmetric AFE vehicle at 90 km on entry when it is traveling at nearly 10 km/s. Although the vehicle does not experience its peak heat transfer at this point, the aerothermal characteristics are still important because they help determine the total heat load and flight trajectory. The conditions at this point are similar to the previous case except that the density is considerably (9 times) lower and the speed is slightly greater.

The first series of plots for this case show the characteristics of the flow on the stagnation streamline. The standoff distance based on the six-fold density increase is 0.167 m or 7.24% of the nose radius. Figure 12 is a semi-log plot of the molar concentrations and Figure 13 is a plot of three temperatures on the stagnation streamline. The translational-rotational temperature reaches a peak of 40,000 K, the vibrational temperature of nitrogen rises to 13,000 K, and the electron temperature peaks at 7200 K. This case is characterized by a high degree of thermal nonequilibrium with a relatively minor excitation of the vibrational and electron modes. As such, we would expect the amount of radiation emitted from the flow to be much lower than at perigee; this is shown in Figure 14. The peak radiative power emitted on the stagnation streamline is 0.0312 W/cm^3, about two orders of magnitude lower than the previous case.

The calculated convective and radiative heating are plotted in Figure 15. The distribution is similar, though about 10% lower, than the 78 km case. The increase in the heat transfer off of the stagnation point is also evident for this case. The stagnation point heat transfer of 15.8 W/cm^2 is similar to the 18.4 W/cm^2 reported by Moss *et al.*[6] The stagnation point radiative heat transfer is 0.112 W/cm^2, which is much smaller than the convective component (164 times less) and the radiative heat transfer for the previous case (53 times less). The much smaller thermal excitation and the lower density of this case has caused the radiative heat transfer to be less significant. Figure 15b also shows that the maximum radiative heat transfer occurs at the stagnation point. There is also a peak of nearly the same magnitude at $\xi \simeq 2.6$ m. The reason for this will become evident when the following contour plots are discussed.

Figure 16 is a contour plot of the N_2 mass concentration and shows that the minimum concentration of diatomic nitrogen occurs, not at the nose, but on the shoulder of the body ($x \simeq 0.75$ m and $y \simeq 2.25$ m). The next three figures are plots of the translational-rotational, N_2 vibrational, and the electron temperatures. The first, Figure 17, shows that the maximum translational-rotational temperature occurs behind the shock on the stagnation point, as expected. However, the peak vibrational and electron temperatures are on the shoulder of the body. This phenomenon is most evident for the electron temperature, which reaches its maximum of nearly 8500 K at this point, as opposed to 7200 K on the stagnation streamline.

The reason why a peak in the degree of chemical reaction and thermal excitation occurs off the stagnation point is made clear by considering Figure 20, which is a streamline plot. Particles are introduced in the free-stream and traced through the flowfield. Consider the streamline that enters the flowfield at $y \simeq 0.9$ m. It passes through an oblique, though strong shock wave, and flows around the body, far enough away from the surface that it does not enter the cool boundary layer. This fluid element experiences rather severe heating inside the entire shock layer and because it travels a significant physical distance, it undergoes appreciable reaction and thermal excitation. Thus, because of the nonequilibrium nature of the flow, the peak degree of reaction can occur off the stagnation point. The final figure is a plot of the radiative emission from the flowfield. The peak radiation power occurs near the shoulder of the body ($x \simeq 0.75$ m and $y \simeq 2.25$ m) where the maximum thermal excitation of the gas has occurred. It should be noted that this effect would be more pronounced for a full scale aerobrake because the gas would have a larger distance to thermalize and would become more radiant.

This case demonstrates that thermo-chemical nonequilibrium can have a large influence on the state of the gas. The largest degree of reaction and thermal excitation do not occur at the stagnation point, and as a result, neither does the peak radiative emission.

Conclusions

The calculation of realistic two-dimensional nonequilibrium flowfields is becoming feasible. In this study the results of an eight-species gas model with six temperatures applied to the Aeroassist Flight Experiment vehicle are presented. They show that the peak convective heat transfer occurs off of the stagnation point. At the perigee conditions of the proposed flight trajectory the radiative heating is about 2 W/cm^2 for most of the aerobrake. However, the radiative heating at the stagnation point is approximately three times larger, which is a significant portion of the total heating to the body. The results at 90 km show that the radiative heating is much less significant for this case. The peak radiative emission occurs off of the stagnation point where the gas has had an appreciable time to thermalize. This effect is due to the strong degree of thermo-chemical nonequilibrium for this combination of free-stream conditions and geometry.

The inclusion of thermal and chemical nonequilibrium effects is mandatory to obtain a realistic description of the flow and to allow radiation to be computed. Also the use of a better vibrational relaxation model than the Landau-Teller model is required because its use results in an overprediction of radiation for the 78 km case by two orders of magnitude.

Appendix

Table A.I. Characteristic Temperatures for Diffusion Model of Vibrational Relaxation

Species	θ_s (K)
N_2	5000
O_2	3350
NO, NO^+	4040

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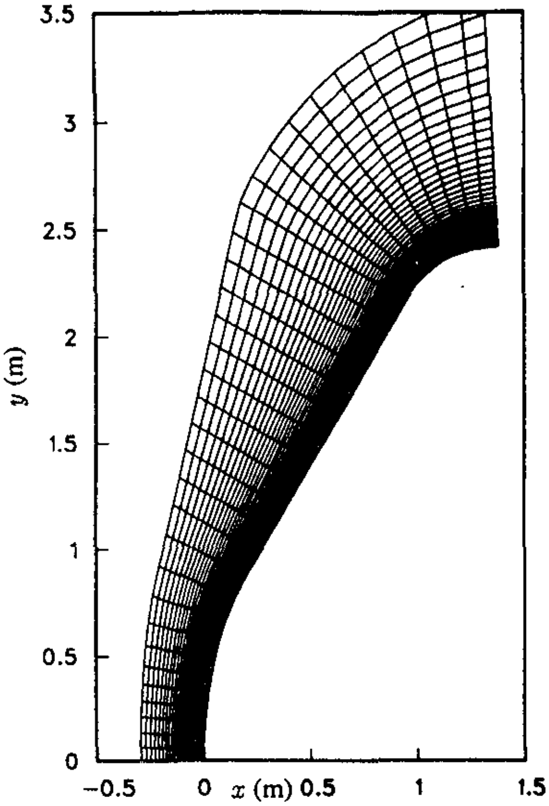


Figure 1. 35×50 grid used for AFE at 78 km.

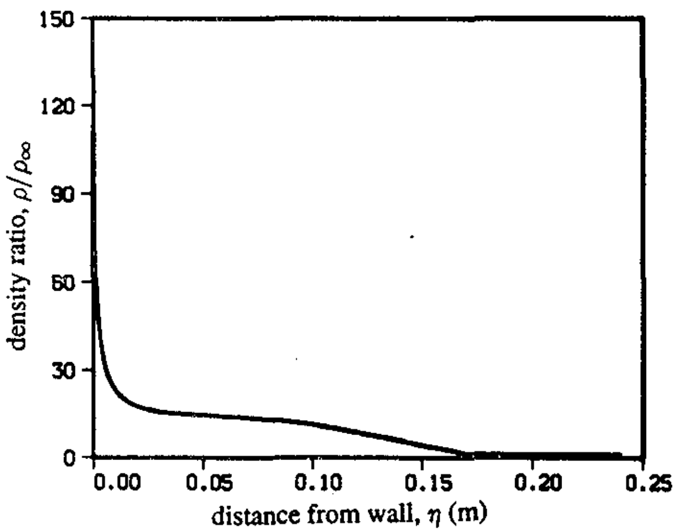


Figure 2. Density ratio on stagnation streamline for AFE at 78 km.

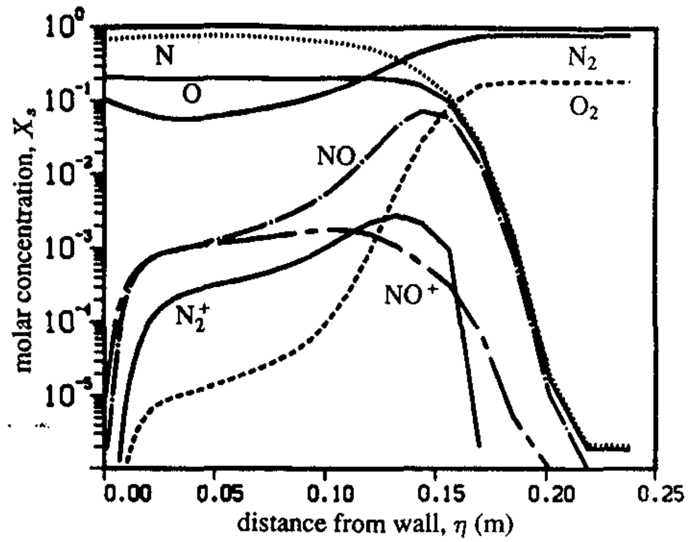


Figure 3. Molar concentrations on stagnation streamline for AFE at 78 km.

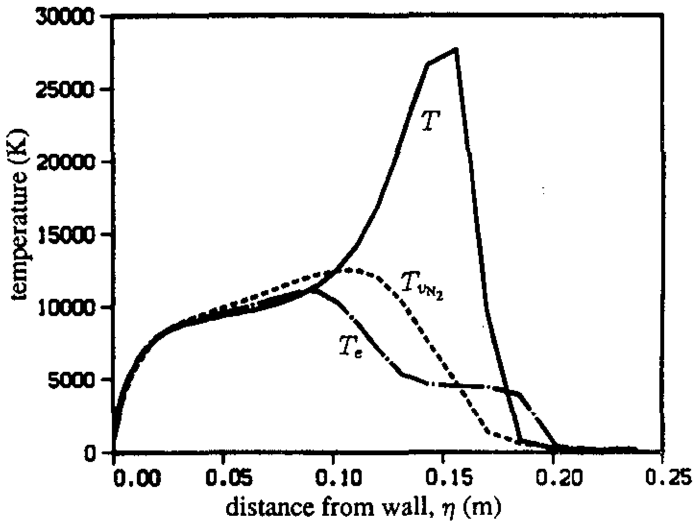


Figure 4. Temperatures on stagnation streamline for AFE at 78 km.

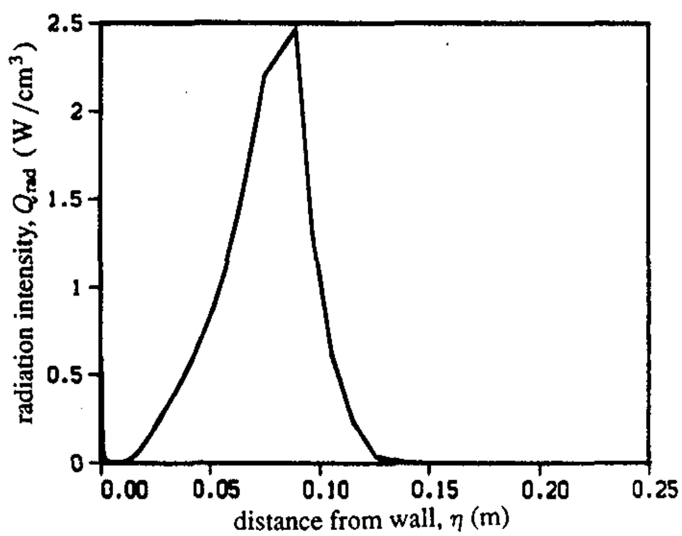
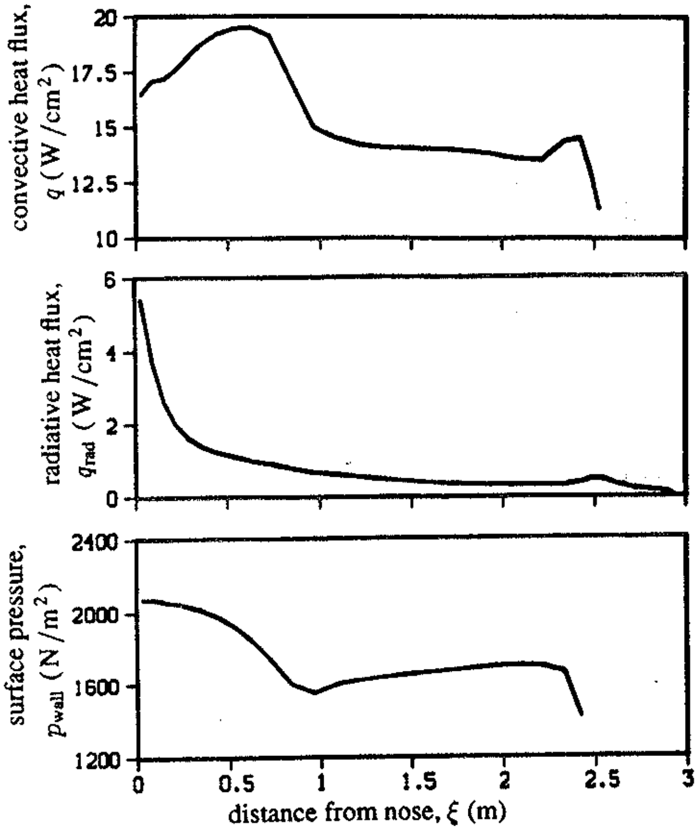


Figure 5. Radiation intensity on stagnation streamline for AFE at 78 km.



Figures 6a, 6b, and 6c. Convective and radiative heat transfer and surface pressure for AFE at 78 km.

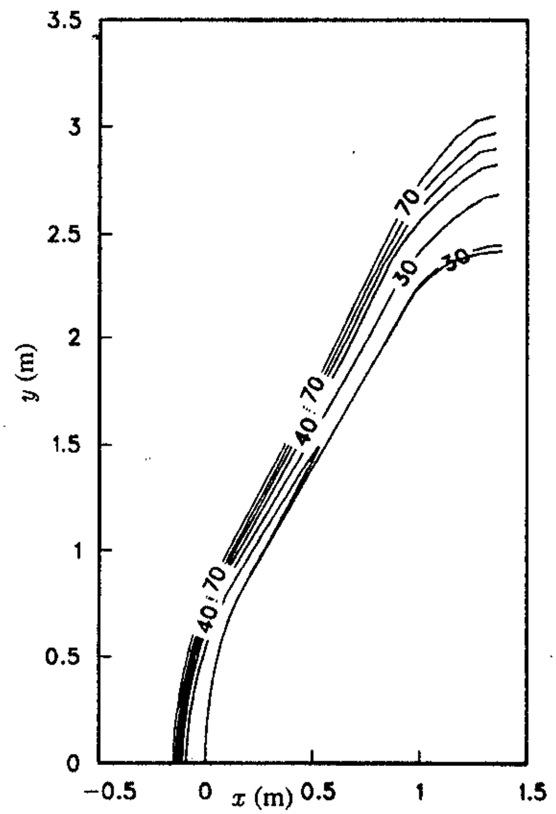


Figure 7. Percent N_2 mass concentration contours for AFE at 78 km.

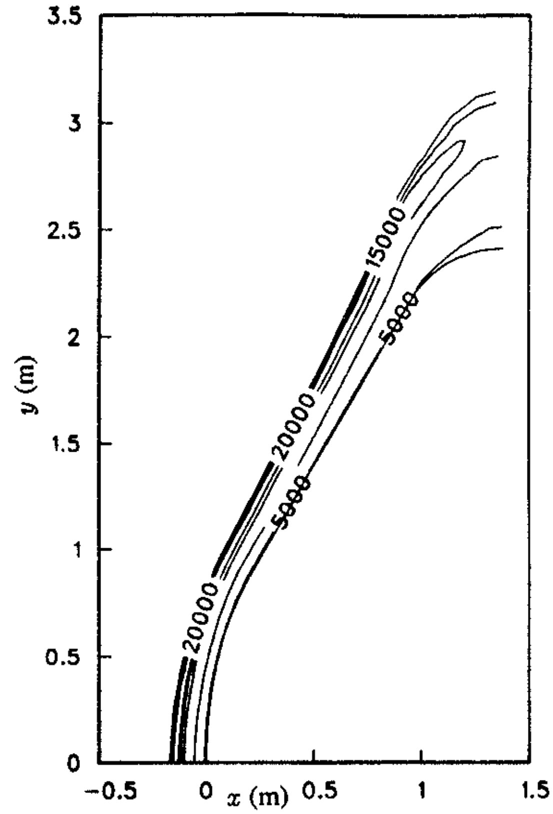


Figure 8. Translational-rotational temperature contours for AFE at 78 km (K).

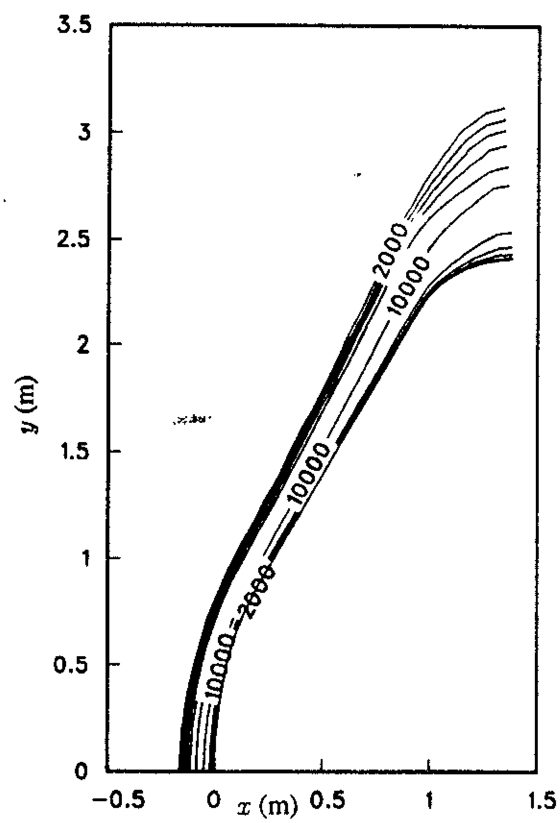


Figure 9. N_2 vibrational temperature contours for AFE at 78 km (K).

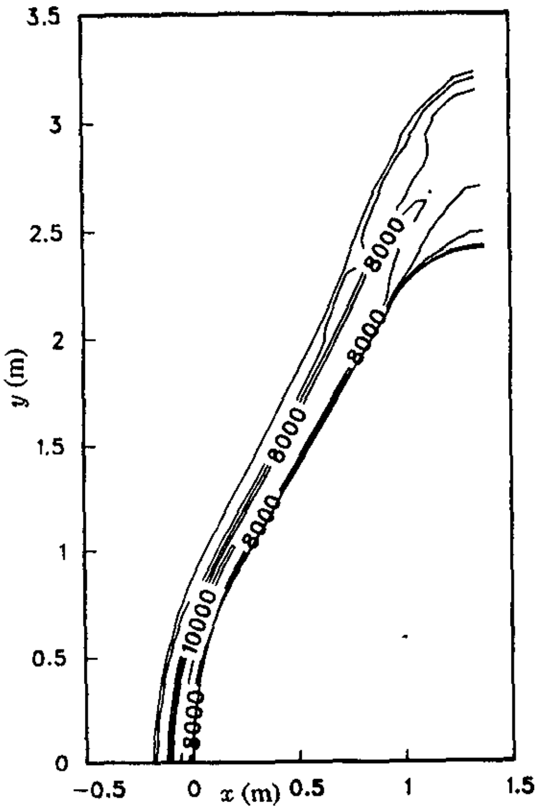


Figure 10. Electron translational temperature contours for AFE at 78 km (K).

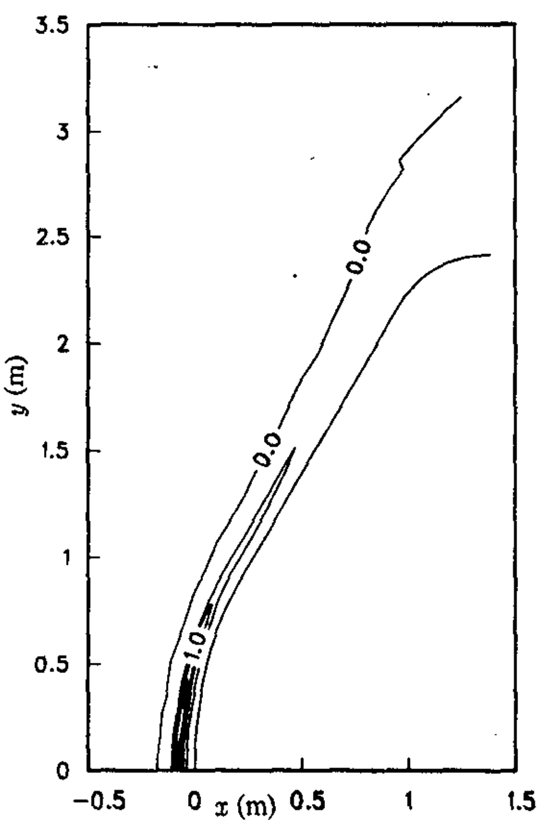


Figure 11. Radiation intensity contours for AFE at 78 km (W/cm^3).

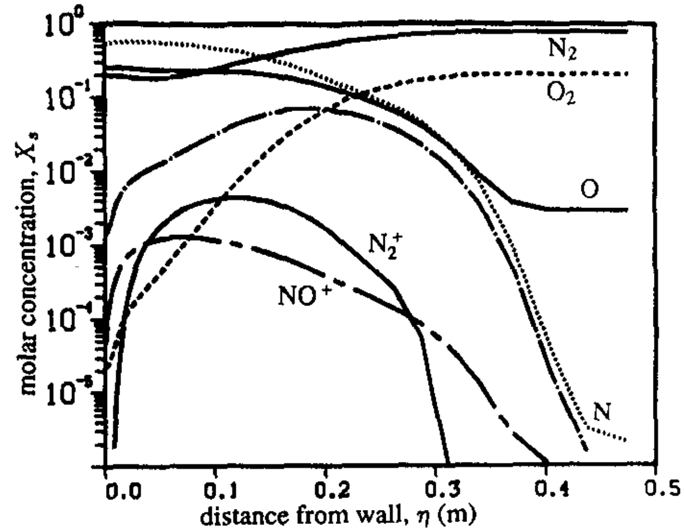


Figure 12. Molar concentrations on stagnation streamline for AFE at 90 km.

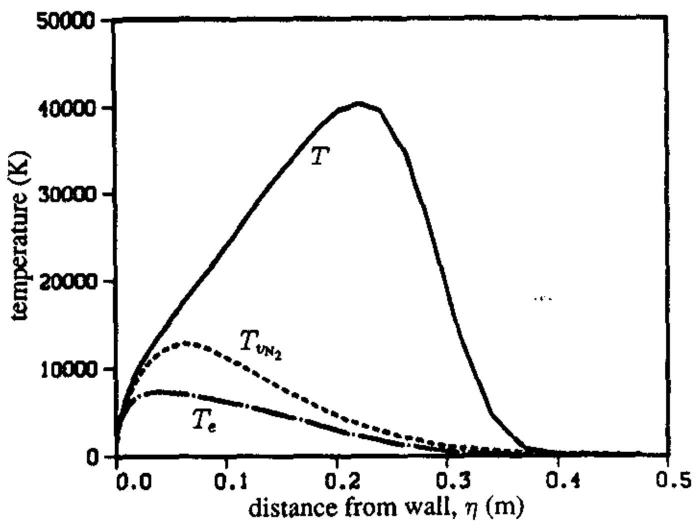


Figure 13. Temperatures on stagnation streamline for AFE at 90 km.

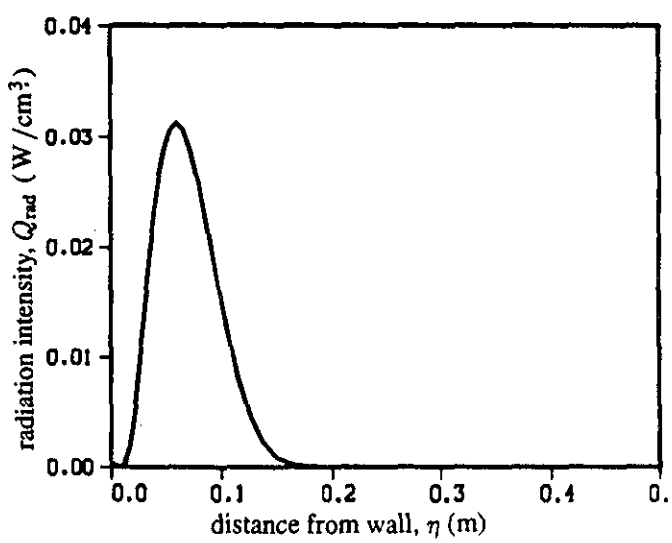
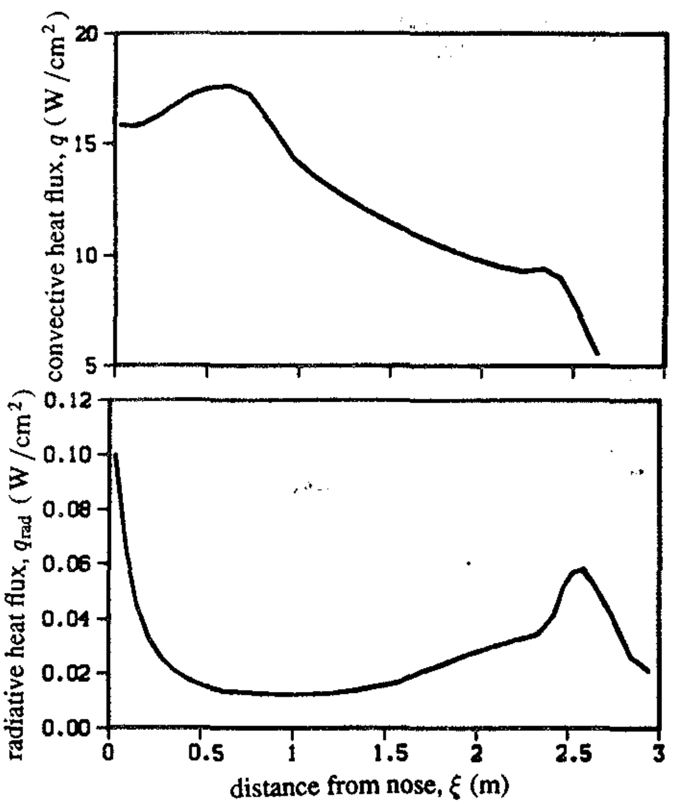


Figure 14. Radiation intensity on stagnation streamline for AFE at 90 km.



Figures 15a and 15b. Convective and radiative heat transfer to AFE at 90 km.

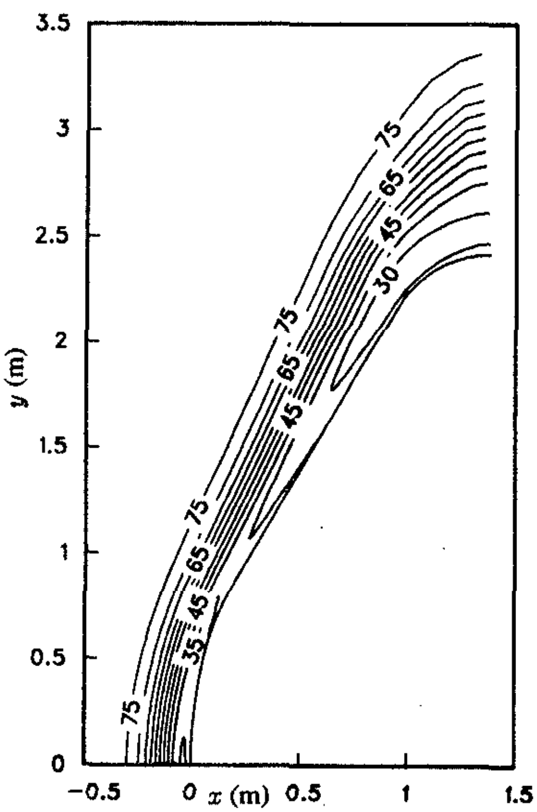


Figure 16. Percent N_2 mass concentration contours for AFE at 90 km.

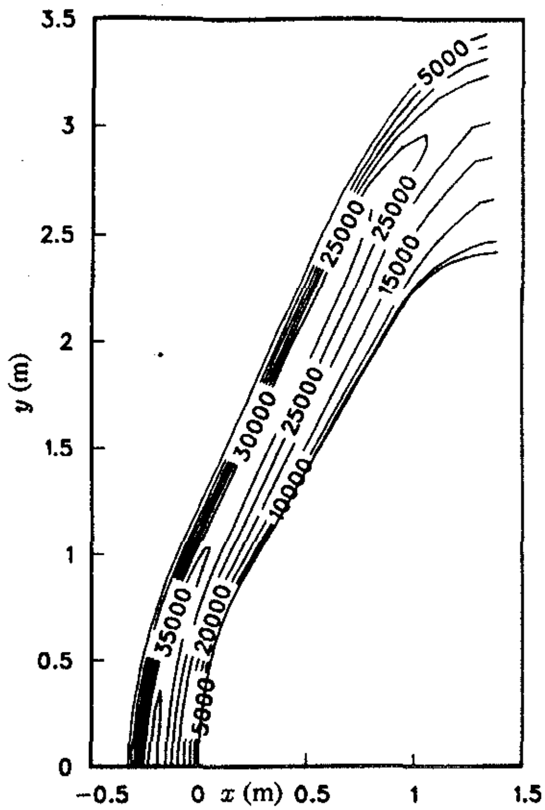


Figure 17. Translational-rotational temperature contours for AFE at 90 km (K).

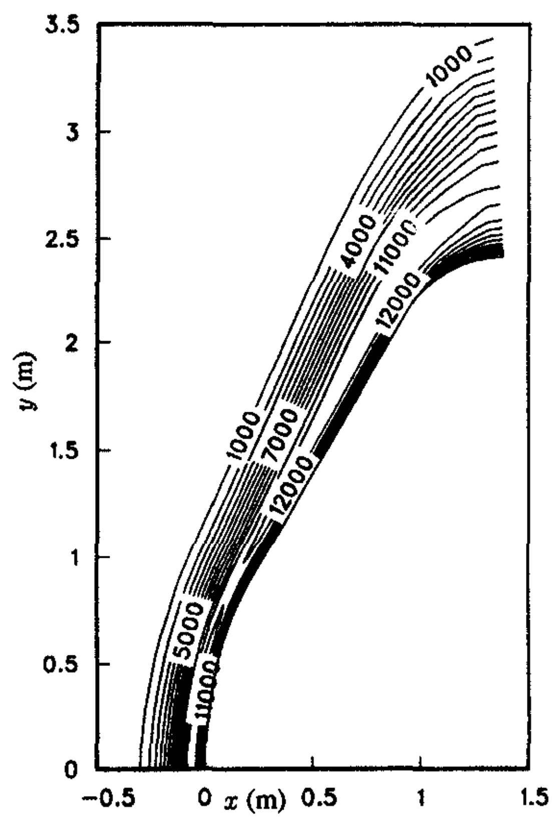


Figure 18. N_2 vibrational temperature contours for AFE at 90 km (K).

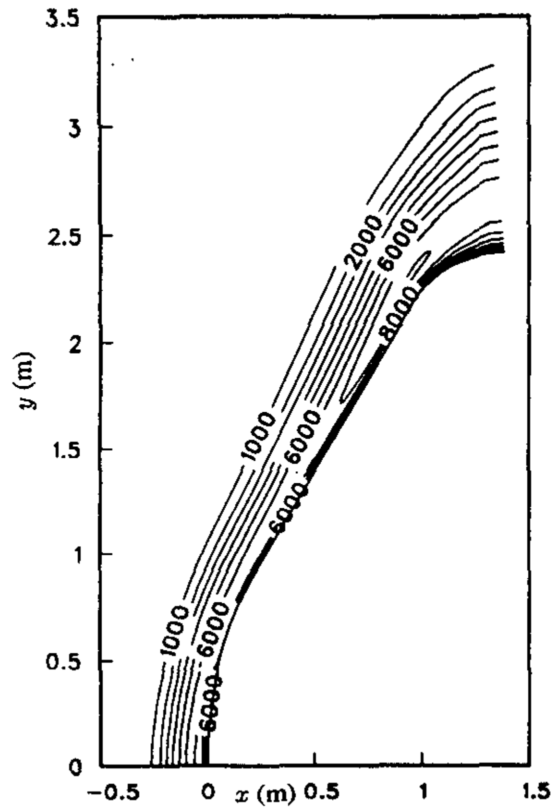


Figure 19. Electron translational temperature contours for AFE at 90 km (K).

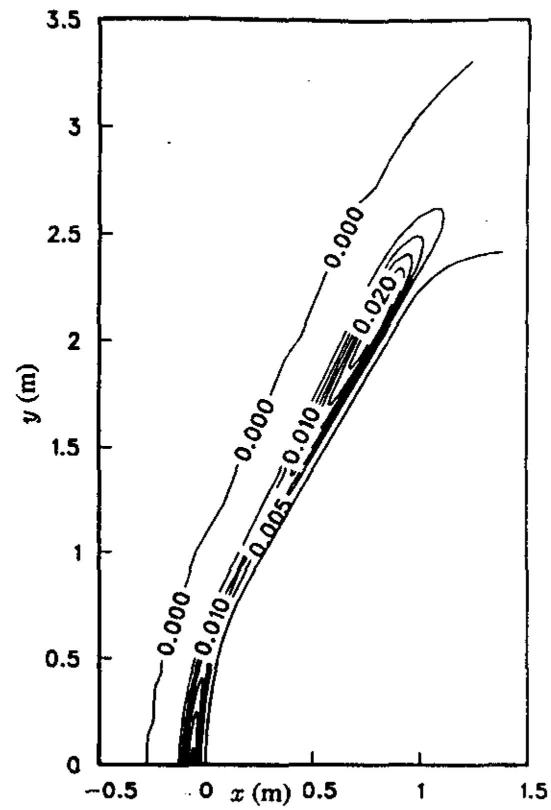


Figure 21. Radiation intensity contours for AFE at 90 km (W/cm^3).

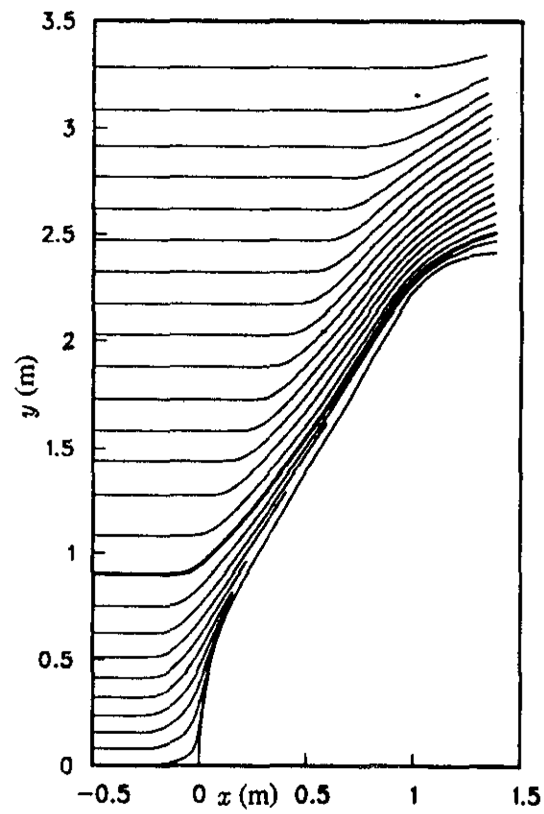


Figure 20. Streamlines for AFE at 90 km.